

Cardiovascular- Pharmacology

At the end of this section students should be able to understand the pharmacology of

- ❖ Anti-hypertensive Agents
- ❖ Anti-angina Drugs
- ❖ Drugs used for the management of heart failure
- ❖ Agents used in the management of Arrhythmia
- ❖ Diuretic agents
- ❖ Agents Used in Cardiac Arrhythmias

Introduction

Blood Pressure

- ❖ Blood pressure is the force exerted by the blood against any unit area of the vessel wall and is usually measured as mmHg.
- ❖ The highest pressure during a heartbeat is systolic pressure (120 mmHg)
- ❖ The lowest point of the pressure is called diastolic pressure and it is approximately 80 mmHg

Introduction...

❖ **Systolic pressure (SP)** is the peak **aortic pressure** that occurs **during the ejection of the blood from** the left ventricle into the aorta during the contraction of the heart.

❖ Factors:

- ✓ stroke volume and the
- ✓ compliance of the arteries and
- ✓ heart rate.

❖ **Diastolic pressure (DP)** is the minimum aortic pressure during relaxation of the heart when there is no blood ejection to the aorta.

❖ Factors:

- ✓ peripheral resistance (TPR)
- ✓ stroke volume and
- ✓ heart rate.

Introduction...

- ❖ Arterial blood pressure (BP) is directly proportionate to the product of the blood flow (cardiac output, CO) and the resistance to passage of blood through pre capillary arterioles (peripheral vascular resistance, PVR):

$$\mathbf{BP = CO \times PVR}$$

- ❖ Physiologically, in both normal and hypertensive individuals, blood pressure is maintained by at three anatomic sites :
 - ✓ Arterioles, post capillary venules (capacitance vessels), and heart.
 - ✓ Kidney contributes to maintenance of blood pressure by regulating the volume of intravascular fluid.
- ❖ Baroreflexes also contributes in maintaining Bp

Introduction...

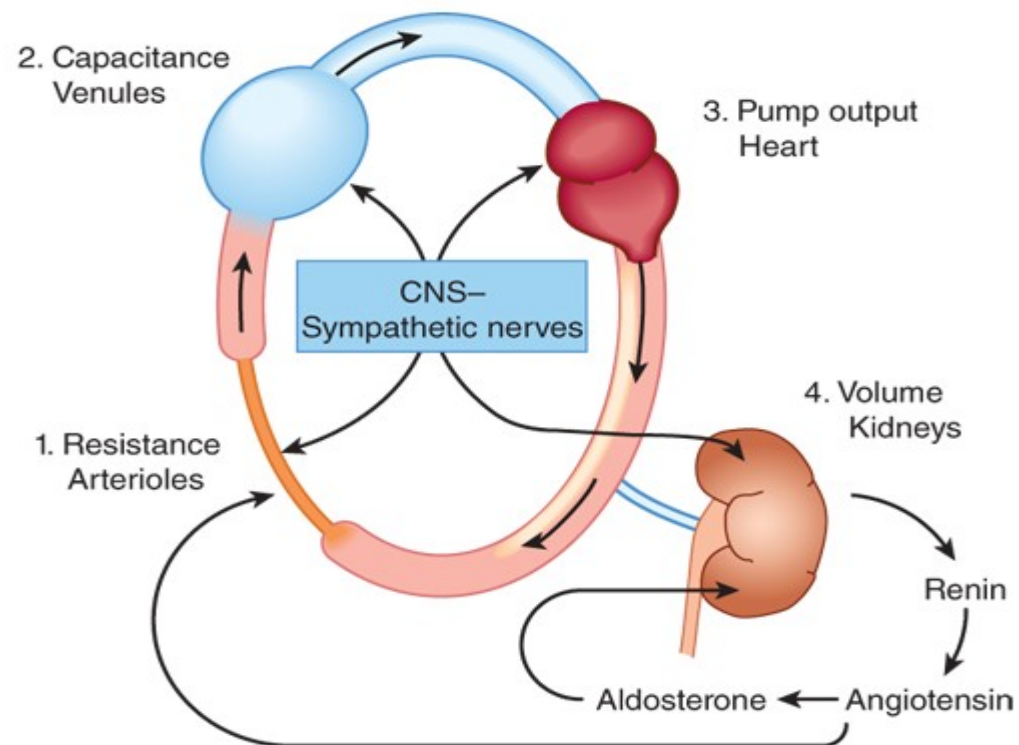


Fig : Anatomic sites of blood pressure control.

Postural Baroreflex

- ❖ Baroreflexes are responsible for rapid, moment-to-moment adjustments in blood pressure, such as in transition from a reclining to an upright posture.
- ❖ Central sympathetic neurons arising from the vasomotor area of the medulla are tonically active.
- ❖ Carotid baroreceptors are stimulated by the stretch of the vessel walls brought about by the internal pressure (arterial blood pressure).
- ❖ Baroreceptor activation inhibits central sympathetic discharge. Conversely, reduction in stretch results in a reduction in baroreceptor activity.

Hypertension

- ❖ Hypertension is the most common cardiovascular disease
- ❖ sustained arterial hypertension damages blood vessels in kidney, heart, and brain and leads to an increased incidence of renal failure, coronary disease, heart failure and stroke.

Classifications of Bp

Classification	Systolic Blood Pressure (mm Hg)		Diastolic Blood Pressure (mm Hg)
Normal	<120	And	<80
Elevated	120-129	And	<80
Stage 1 hypertension	130-139	or	80-89
Stage II hypertension	≥ 140	Or	≥ 90

Terminologies

Gestational hypertension – Gestational hypertension refers to elevated blood pressure first detected **after 20 weeks** of gestation in the absence of **proteinuria** or other diagnostic features of preeclampsia

Chronic (preexisting) hypertension – Chronic hypertension is defined as hypertension that antedates pregnancy, **is present before the 20th week of pregnancy**, or persists **longer than 12 weeks** postpartum

Preeclampsia: refers to the syndrome of **new onset** of hypertension and **proteinuria** or new onset of hypertension and end-organ dysfunction with or without **proteinuria**

- ✓ most often after **20 weeks** of gestation in a previously normotensive woman
- ✓ **Eclampsia** is diagnosed when seizures have occurred

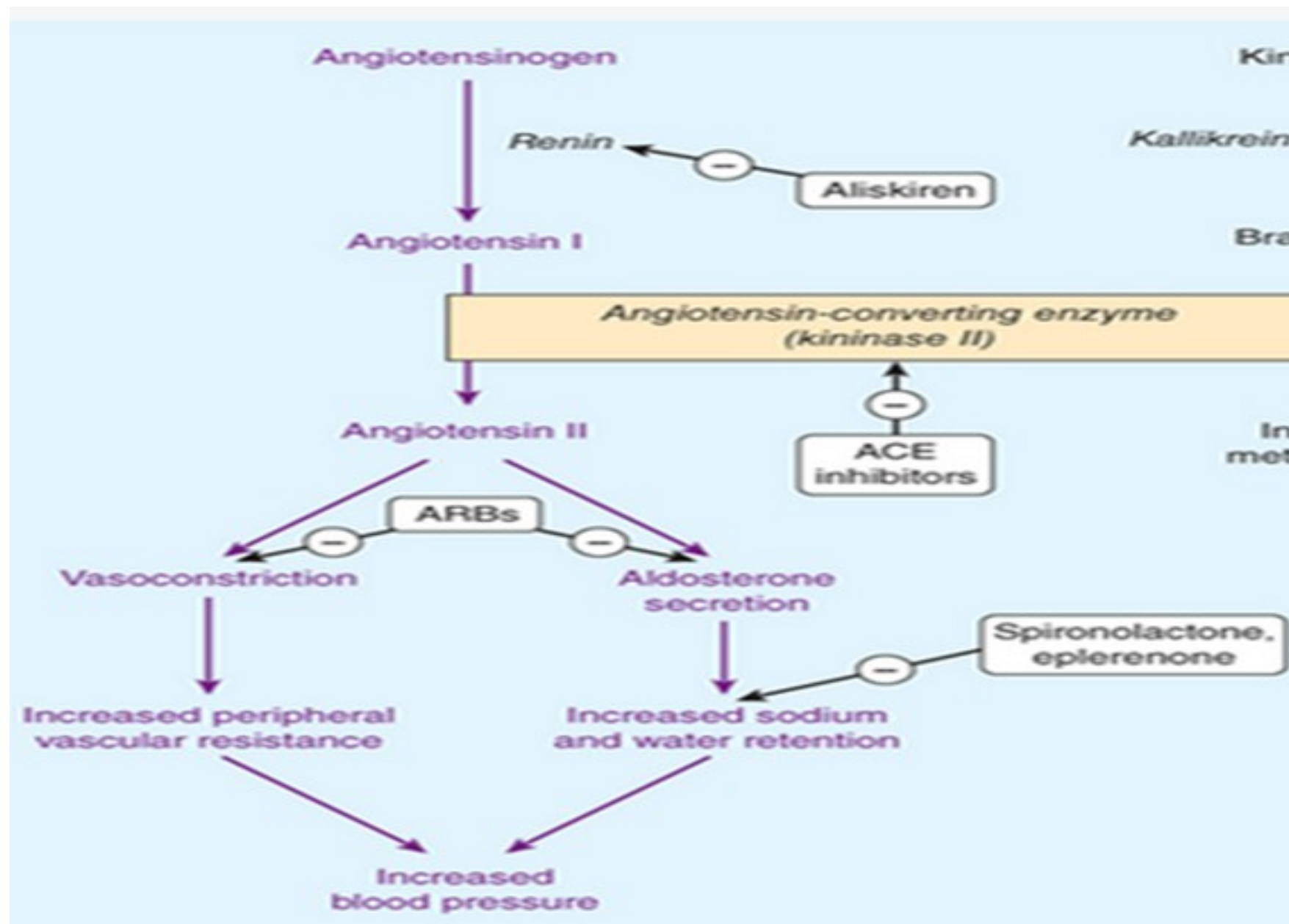
Etiology of Hypertension

- ❖ A specific cause of hypertension can be established in only 10–15% of patients.
- ❖ Based on the either the etiology is known or not known hypertension is classified as
 - ✓ Primary (essential) hypertension..... no specific cause is known
 - ✓ secondary hypertension..... Specific cause known
- ✓ In most cases, elevated blood pressure is associated with an overall increase in **resistance to flow** of blood through arterioles, whereas **cardiac output** is usually normal.

Renal Response to Decreased Blood Pressure

- ❖ By controlling blood volume, the kidney is primarily responsible for long-term blood pressure control.
- ❖ A **reduction in renal perfusion** pressure causes intra **renal redistribution** of blood flow and increased reabsorption of salt and water.
- ❖ In addition, decreased pressure in renal arterioles as well as sympathetic neural activity (via β adrenoceptors) stimulates production of renin, which increases production of angiotensin II, **Angiotensin II causes** Direct constriction of resistance vessels

- ✓ Stimulation of **aldosterone** synthesis in the **adrenal cortex**, which increases renal sodium absorption and intravascular blood volume.
- ❖ Vasopressin released from the **posterior pituitary gland** also plays a role in maintenance of blood pressure through its ability to regulate water reabsorption by the kidney



Basic Pharmacology of Antihypertensive Agents

❖ antihypertensive agents act at one or more of the four anatomic control sites

❖ Classifications

✓ *Diuretics*

Reduce blood pressure and volume by depleting the body of sodium and reducing blood volume and perhaps by other mechanism.

✓ *Sympathoplegic agent*

lower blood pressure by reducing **peripheral vascular resistance** and inhibiting **cardiac function**.

✓ *Direct vasodilators*

Reduce pressure by relaxing vascular smooth muscle, thus dilating resistance vessels and—to varying degrees—increasing capacitance as well.

✓ **Agents that block production or action of angiotensin** and thereby reduce peripheral vascular resistance

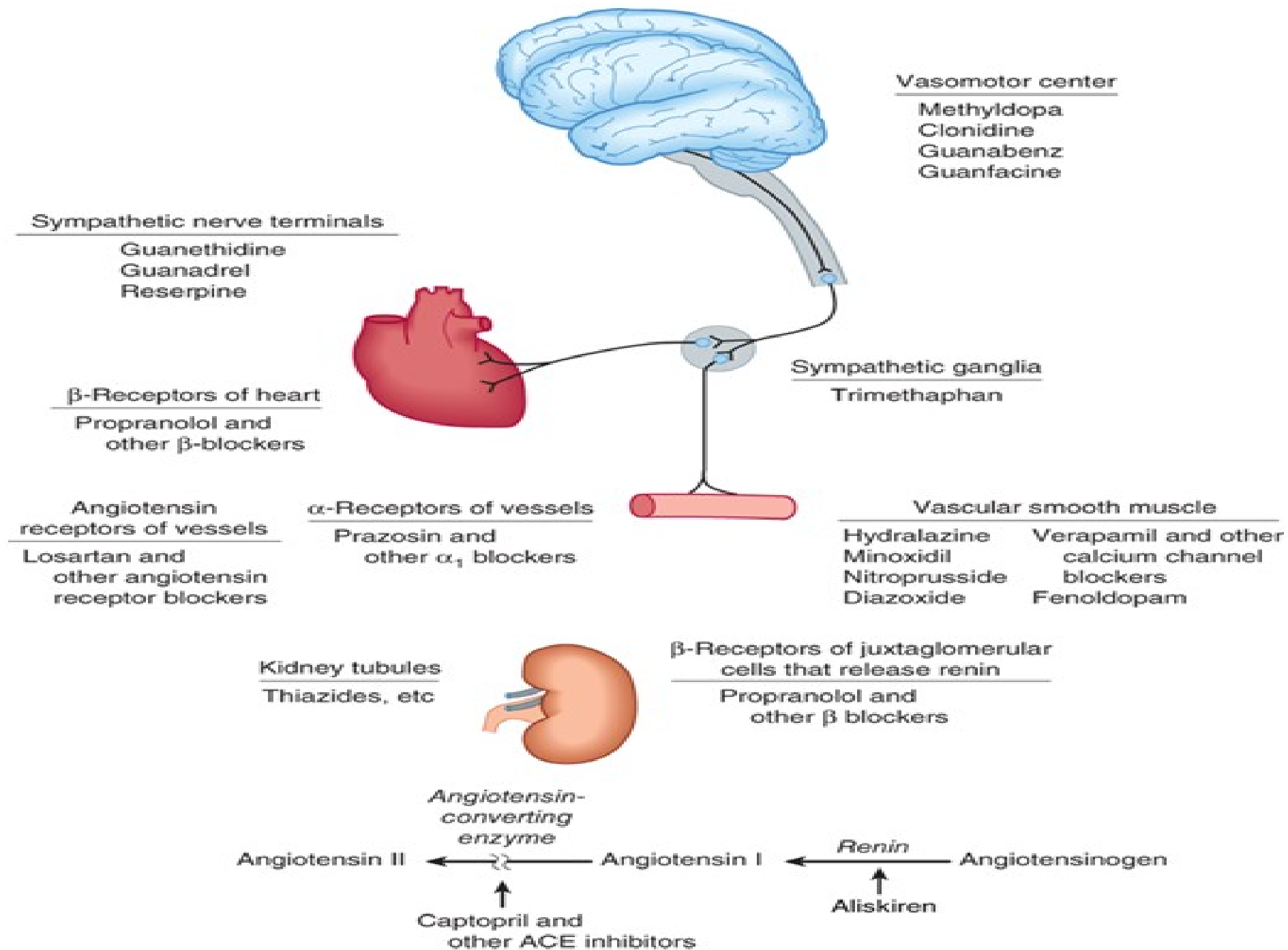


Fig: Sites of action of the major classes of antihypertensive drugs.

Diuretics

❖ Mechanism of action

- ✓ lower blood pressure primarily by depleting body sodium stores.

Sodium is believed to contribute to **vascular resistance** by increasing vessel stiffness and neural reactivity.

- ✓ Initially they ↓ BP by reducing **blood volume** and CO how ever, peripheral vascular resistance may increase.
- ✓ After 6–8 weeks, **cardiac output** returns toward normal while peripheral vascular resistance declines

- ❖ lowering BP by 10–15 mm Hg (used for mild-moderate HTN alone and in combination with other agents for severe HTN).

Types of diuretics and their role in HTN

- ❖ **Thiazide diuretics** –For **mild** or **moderate** hypertension and normal renal and cardiac function They result in potassium depletion .
- ❖ **Potassium-sparing** diuretics are useful both to avoid excessive potassium depletion and to enhance the natriuretic effects of other diuretics.
- ❖ **Aldosterone receptor antagonists:** have a favorable effect on cardiac function in people with heart failure.

Diuretics

- ❖ When the dose of
- ✓ Loop diuretics is increased BP response continues to increase at doses many times greater than the usual therapeutic dose but not for **thiazide** diuretics.

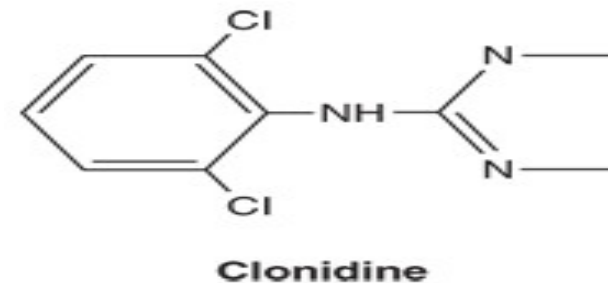
Drugs That alter Sympathetic Nervous System Function

❖ Mechanism of action

- ✓ Reduce sympathetic outflow from **vasomotor centers** in the brain stem.
- ✓ **Drugs:-** Clonidine, Methyldopa, Guanabenz and guanfacine.

Clonidine:-

- ❖ Clonidine Controls Bp by the following effect
 - ✓ Reduce CO due to decreased heart rate
 - ✓ relaxation of capacitance vessels,
 - ✓ reduction in peripheral vascular resistance.



Pharmacokinetics & Dosage

- ❖ Clonidine is lipid-soluble and rapidly enters the brain from the circulation.
- ❖ Because of its relatively short half-life oral clonidine must be **given twice** a day.
- ❖ When applied as transdermal patch, clonidine **reduces blood pressure** for **7** days after a single dose.
 - ✓ This preparation appears to produce **less sedation** than clonidine tablets but may be associated with local skin reactions.

Side effects

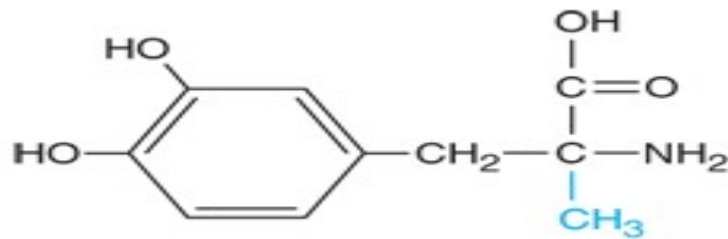
- ✓ Dry mouth and
- ✓ Sedation

Contra-Indications

- ✓ Patients who are at risk for mental depression
- ✓ Patients with tricyclic antidepressants

Methyldopa (L- α -methyl-3,4-dihydroxy phenylalanine)

- ❖ Methyldopa is an analog of L-dopa and is converted to
 - ✓ α -methyldopamine and
 - ✓ α -methylnorepinephrine.



- ❖ Alpha-methylnorepinephrine is stored in adrenergic nerve vesicles, where it replaces norepinephrine, and is released by **nerve stimulation** to interact with postsynaptic adrenoceptors.
- ❖ Methyldopa's antihypertensive action is due to **stimulation of central α adrenoceptors** by α methylnorepinephrine or α -methyldopamine.

Pharmacokinetics & Dosage

- ❖ Methyldopa enters the brain via an **aromatic amino** acid transporter.
- ❖ The usual oral dose of methyldopa produces its maximal antihypertensive effect in **4–6 hours**, and the effect can persist for up to **24 hours**.
- ❖ The action persists after the parent drug has disappeared from the circulation

Toxicity

- ✓ Sedation
 - ✓ Impaired mental concentration,
 - ✓ Nightmares,
 - ✓ Mental depression,
 - ✓ Vertigo,
 - ✓ Lactation.
- ❖ Guanabenz and guanfacine are centrally active antihypertensive drugs that share the central α -adrenoceptor-stimulating effects of clonidine.
 - ✓ They do not appear to offer any **advantages over clonidine** and are rarely used.

Adrenergic Neuron-blocking Agents

- ❖ These drugs lower blood pressure by preventing normal physiologic **release** of **norepinephrine** from postganglionic sympathetic neurons.
- ❖ **Guanethidine:**
 - ❖ **MOA:** Guanethidine inhibits the release of norepinephrine from sympathetic nerve endings.
 - ✓ Has long half-life (5 days) and the onset of sympathoplegia is gradual (maximal effect in **1–2 weeks**), and sympathoplegia persists for a comparable period after cessation of therapy.
 - ✓ The dose should not ordinarily be increased at intervals shorter **than 2 weeks**.
 - ✓ Common side effects include marked **postural hypotension, diarrhea, and impaired ejaculation**.
 - ✓ Because of these **adverse effects**, **guanethidine** is now rarely used.

Reserpine

MOA:

❖ Reserpine blocks the ability of aminergic transmitter vesicles to **take up** and **store biogenic amines**, by interfering with the vesicular membrane-associated transporter (VMA).

❖ Dosage and PK of reserpine

✓ Half life=24–48hrs

✓ Bio-availability= 50 %

✓ Dose= 0.25 mg/d

❖ Toxicity

At low doses-----little postural hypotension, EPS

At high dose-----sedation, **lassitude**, nightmares, and severe **mental depression** (stop medication if it occurs), **mild diarrhea** and **gastrointestinal cramps** and increases **gastric acid secretion**

❖ C/I

PUD

Adrenoceptor Antagonists

- ❖ Of the large number of β blockers tested, most have been shown to be effective in lowering blood pressure.
- ❖ All β -adrenoceptor-blocking agents are useful for lowering blood pressure in **mild to moderate** hypertension.
- ❖ In severe hypertension, β blockers are especially useful in preventing the **reflex tachycardia** that often results from treatment with direct vasodilators.
- ❖ Beta blockers have been shown to reduce mortality after in **myocardial infarction** and in patients with **heart failure**.
- ❖ Drugs in this class: propranolol, metoprolol & atenolol, nadolol, carteolol, betaxolol, & bisoprolol, labetalol, carvedilol, & nebivolol, esmolol.

Propranolol

- ❖ Propranolol was the first β blocker shown to be effective in hypertension and ischemic heart disease.
- ❖ MOA:
 - ✓ Decrease cardiac output
 - ✓ Inhibits the stimulation of renin production by catecholamines (mediated by β_1 receptors).
 - ✓ Reduce peripheral presynaptic β adrenoceptors to reduce sympathetic vasoconstrictor nerve activity.

Propranolol...

❖ Toxicity

- ✓ Bradycardia (Due to β_1 - blocking agent)
- ✓ Asthma (β_2 -blocking action)

Metoprolol & Atenolol

- ❖ Metoprolol = propranolol (in β_1 adrenergic blockage) and **50- to 100-fold** less potent than propranolol in blocking β_2 receptors (make it good for people with asthma and diabetes)
- ❖ Metoprolol is extensively metabolized by CYP2D6 with high first-pass metabolism ($t_{1/2}$ =4-6 h).
- ❖ Atenolol
 - ✓ Not metabolized ($t_{1/2}$ = 6h and once daily dose 50-100 mg/d)
 - ✓ Atenolol is reported to be **less effective** than metoprolol in preventing the complications of hypertension
 - ✓ Patients with reduced renal function should receive lower doses.

Nadolol, Carteolol, Betaxolol, & Bisoprolol

❖ Nadolol and carteolol (Non-selective β bloker)

- ✓ Not metabolized and are excreted to a considerable extent in the urine.
- ✓ Patients with reduced renal function should receive correspondingly reduced doses of nadolol and carteolol
- ✓ Dose = Nadolol is begun at a dosage of 40 mg/d and carteolol at 2.5 mg/d.

❖ Betaxolol and bisoprolol (β 1-selective blockers)

- ✓ Metabolized in the liver but have long half-lives.
- ✓ Dose = betaxolol 10 mg/d and
bisoprolol at 5 mg/d

Pindolol, Acebutolol, & Penbutolol

- ❖ These drugs are partial agonists, ie, β blockers with some intrinsic sympathomimetic activity.
- ❖ They lower blood pressure but are rarely used in **hypertension**

Labetalol, Carvedilol, & Nebivolol

- ❖ These drugs have both β -blocking and **vasodilating** effects.
- ❖ **Labetalol** Blood pressure is lowered by reduction of systemic **vascular resistance** (via α blockade) **without significant alteration** in heart rate **or cardiac output**.
 - ✓ Because of its combined α - and β -blocking activity, labetalol is useful in treating the hypertension of **pheochromocytoma** and **hypertensive emergencies**.
Dose =
 - ✓ Oral = 200 to 2400 mg/d
 - ✓ IV bolus = 20–80 mg for hypertensive emergencies
- ❖ **Carvedilol** for ordinary hypertension is
 - ✓ Initial dose = 6.25 mg twice daily.
 - ✓ reduces mortality in patients with **heart failure**
- ❖ **Nebivolol**
 - ✓ is a β_1 -selective blocker with **vasodilating** properties that are not mediated by α blockade.
 - ✓ Nebivolol is **extensively metabolized** and has active metabolites ($t_{1/2}$ = 10-12 hrs)

Starting dose = 5 mg/d – 40 mg/d.

Esmolol

- Esmolol is a β_1 -selective blocker
- metabolized via hydrolysis by red blood cell esterases ($t_{1/2}$ =9–10 minutes)
- is administered by intravenous infusion.
- is used for management of **intraoperative** and **postoperative hypertension**

Alpha1 Blockers (Prazosin, terazosin, and doxazosin)

- These agents **produce less reflex tachycardia** when lowering blood pressure than do **nonselective α antagonists such as phentolamine**.
- The drugs are more effective when used in combination with other agents, such as a **β blocker** and a **diuretic**, than when used alone
- Useful with **Prostatic hyperplasia** and **bladder** obstruction symptoms with HTN.

Alpha1 Blockers (Prazosin, terazosin, and doxazosin)

- ❖ **Dose** = Terazosin can often be **given once daily**, with doses of 5–20 mg/d.
- ❖ Doxazosin is usually given once daily starting at **1 mg/d** and progressing to **4 mg/d**
- ❖ **MOA** =
 - ✓ Blocking α_1 receptors in arterioles and venules
 - ✓ Alpha blockers reduce arterial pressure by dilating both **resistance** and **capacitance vessels**
- ❖ **Toxicity**
 - Retention of **salt and water** occurs when these drugs are administered without a diuretics.

Vasodilators

➤ Drugs in this class includes:

- ✓ **Oral vasodilators:** minoxidil, hydralazine
- ✓ **Parenteral vasodilators,** nitroprusside and fenoldopam, which are used to treat hypertensive emergencies;
- ✓ **Calcium channel blockers**

Hydralazine



- ✓ Dilates only arteries.
- ✓ The combination of hydralazine with nitrates is used for **heart failure** with **HTN**.

Dose =

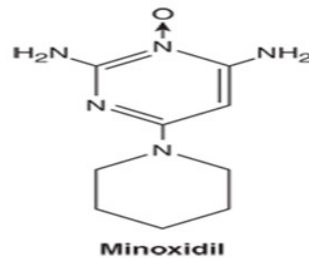
40 to 200 mg/d.

failur

Toxicity

- The most common adverse effects of hydralazine are headache, **nausea, anorexia, palpitations, sweating, and flushing**

Minoxidil



➤MOA

- ✓ Potassium opener (Increased potassium permeability stabilizes the membrane at its resting potential and makes contraction less likely)
- ✓ **Dialtes only** arterioles
- minoxidil should replace hydralazine when maximal doses of the latter are not effective or in patients with **renal failure** and **severe hypertension**, who do not respond well to hydralazine.

TOXICITY

- **Tachycardia, palpitations, angina, and edema** if co-administered with β blockers and diuretics are inadequate.
- **Headache, sweating and hypertrichosis.**

Nitroprusside

➤ Sodium nitroprusside is a powerful parenterally administered vasodilator used for

- ✓ hypertensive emergencies
- ✓ severe heart failure.

➤ Dilates both **arterial** and **venous** vessels,

❖ .MOA

➤ Activate guanyl cyclase,



➤ Increase intracellular cGMP,



➤ Relaxes vascular smooth muscle

Pharmacokinetics & Dosage

- **Nitroprusside** is a complex of iron, cyanide groups, and a **nitroso moiety**.
- Metabolized by uptake into red blood cells with release of nitric oxide and cyanide.
- Cyanide in turn is metabolized by the mitochondrial enzyme rhodanese, in the presence of a sulfur donor, to the less toxic **thiocyanate**.
- Thiocyanate is distributed in extracellular fluid and slowly **eliminated** by the kidney.
- Effects of Nitroprusside disappear within 1–10 minutes after discontinuation.

❖ Toxicity

- The most serious toxicity is related to accumulation of cyanide; metabolic acidosis, **arrhythmias**, excessive **hypotension**, and death have resulted.
- **Sodium thiosulfate** and **Hydroxocobalamin** are used for the management of cyanide toxicity

Diazoxide

- ❖ **Diazoxide** is an effective and relatively long-acting potassium channel opener that causes hyperpolarization.
- ❖ Has same chemical structure with diuretics but do not have **any diuresis activity**
- ❖ Used to treat **hypertensive emergencies**.
- ❖ **Non-cardiovascular action**
 - ✓ Inhibits insulin release from the **pancreas** (used orally in the USA for the treatment of **hypoglycemia in hyperinsulinism**)

Pharmacokinetics & Dosage

- ❖ $t_{1/2} = 24$ hours,
- ❖ Onset = 5 minutes
- ❖ Duration = 4–12 hours.
- ❖ The hypotensive effects will be greater when patients are pretreated with β blockers to prevent the **reflex tachycardia** and associated increase in cardiac output.
- ❖ **Toxicity**
 - ✓ Hypotension (may result in angina and MI)
 - ✓ Hyperglycemia

Fenoldopam

- ❖ It is a peripheral **arteriolar dilator** used for
 - ✓ hypertensive emergencies and
 - ✓ postoperative hypertension.

MOA

- ❖ It acts primarily as an agonist of dopamine D1 receptors, resulting in dilation of peripheral arteries
- ❖ **Pharmacokinetics**
- ❖ Fenoldopam is rapidly metabolized, primarily by **conjugation**.
- ❖ Its half-life is 10 minutes.
- ❖ Fenoldopam is initiated at a low dosage (0.1 mcg/kg/min).

Toxicity

- ❖ The major toxicities are
 - ✓ reflex tachycardia,
 - ✓ headache,
 - ✓ flushing.
 - ✓ **glaucoma**.

Calcium Channel Blockers

- ❖ Calcium channel blockers reduce peripheral resistance and blood pressure.
- ❖ **CCB** are orally active agents and are characterized by
 - high first-pass effect,
 - high plasma protein binding, and
 - extensive metabolism.
- ❖ **Verapamil** and *diltiazem* are also used by the intravenous route.
- ❖ The mechanism of action in hypertension (and, in part, in angina) is inhibition of calcium influx into arterial smooth muscle cells

Classified in to two

- ❖ Dihydropyridine family (amlodipine, felodipine, isradipine, nicardipine, nifedipine, nisoldipine, and nitrendipine)
- ❖ Non- dihydropyridine (Verapamil, diltiazem)

Difference between them

- ✓ dihydropyridine are more selective as vasodilators and have less cardiac depressant effect than verapamil and diltiazem
- ✓ Binding site

Calcium Channel Blockers

Type	Channel Name	Where Found	Properties of the Calcium Current	Blocked By
L	Ca _v 1.1-Ca _v 1.4	Cardiac, skeletal, smooth muscle, neurons (Ca _v 1.4 is found in retina), endocrine cells, bone	Long, large, high threshold	Verapamil, DHPs, Cd ²⁺ , ω-aga-IIIa
T	Ca _v 3.1-Ca _v 3.3	Heart, neurons	Short, small, low threshold	sFTX, flunarizine, Ni ²⁺ (Ca _v 3.2 only), mibefradil ¹
N	Ca _v 2.2	Neurons, sperm ²	Short, high threshold	Ziconotide, ³ gabapentin, ⁴ ω-CTXGVIA, ω-aga-IIIa, Cd ²⁺
P/Q	Ca _v 2.1	Neurons	Long, high threshold	ω-CTX-MVIIc, ω-aga-IVA
R	Ca _v 2.3	Neurons, sperm ²	Pacemaking	SNX-482, ω-aga-IIIa

Properties of several voltage-activated calcium channels.

Pharmacodynamics of CCB

- ❖ The voltage-gated L type is the dominant type of calcium channel in **cardiac** and **smooth muscle** and is known to contain several drug receptors.
- ❖ It consists of $\alpha 1$ (the larger, pore-forming subunit), $\alpha 2$, β , γ , and δ subunits.
 - ✓ Nifedipine and other dihydropyridines have been demonstrated to bind to one site on the $\alpha 1$ subunit,
 - ✓ whereas verapamil and diltiazem appear to bind to closely related but not identical receptors in **another region of the same subunit**.
- ❖ Binding of the drug reduces the frequency of opening in response to depolarization
 - ✓ decrease in transmembrane calcium current.... results in long-lasting relaxation

Toxicities of CCB

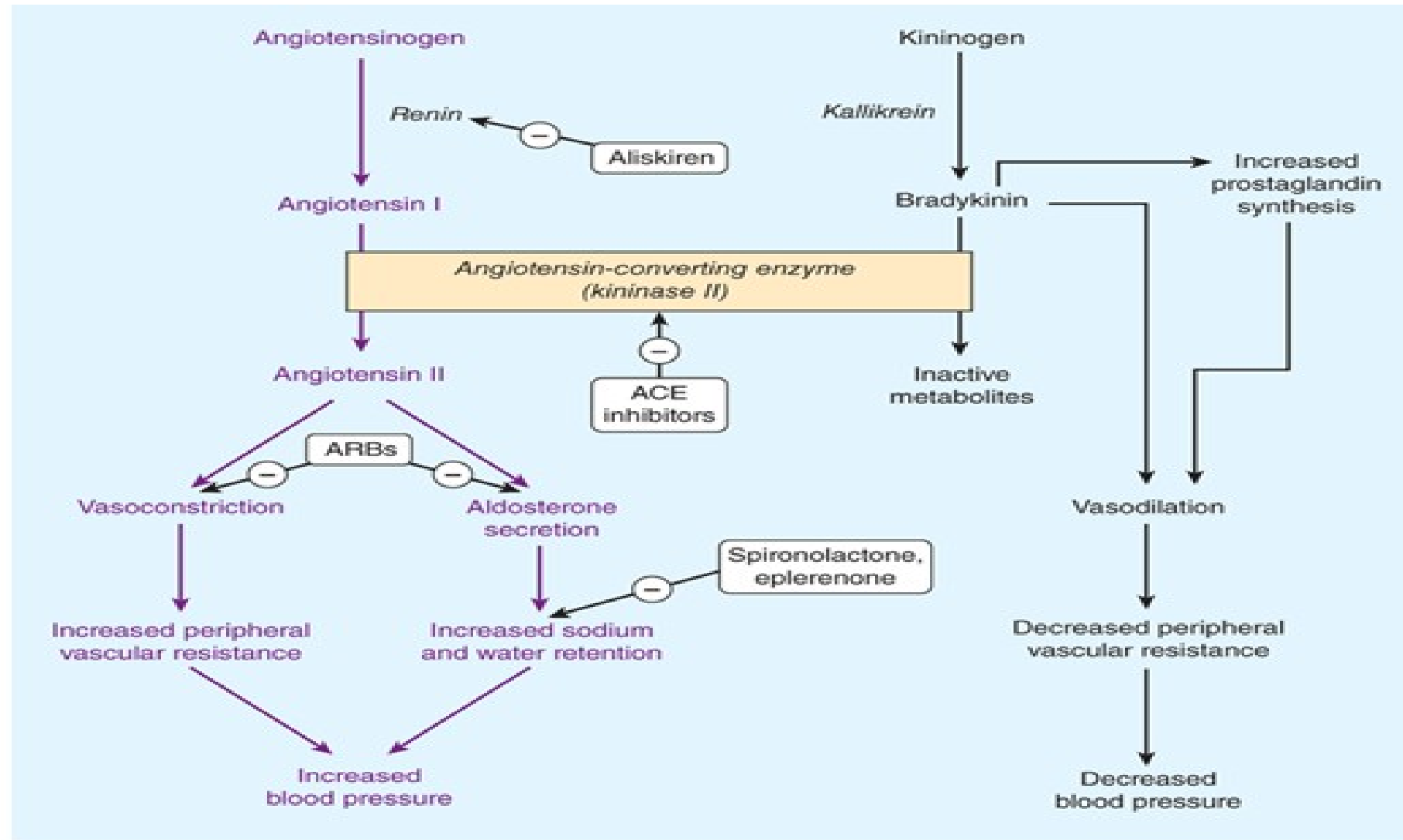
- ❖ The most important toxic effects reported for calcium channel blockers are direct extensions of their therapeutic action
- ❖ Excessive inhibition of calcium influx can cause serious cardiac depression, including
 - bradycardia,
 - atrioventricular block,
 - cardiac arrest, and heart failure.

Inhibitors of Angiotensin...

- ❖ Renin release from the kidney cortex is stimulated by
 - ✓ Reduced renal arterial pressure,
 - ✓ Sympathetic neural stimulation, and
 - ✓ Reduced sodium delivery or increased sodium concentration at the distal renal tubule.
- ❖ Angiotensin II has vasoconstrictor and sodium-retaining activity.

Inhibitors of Angiotensin

❖ MOA



Angiotensin-converting Enzyme (ACE) inhibitors

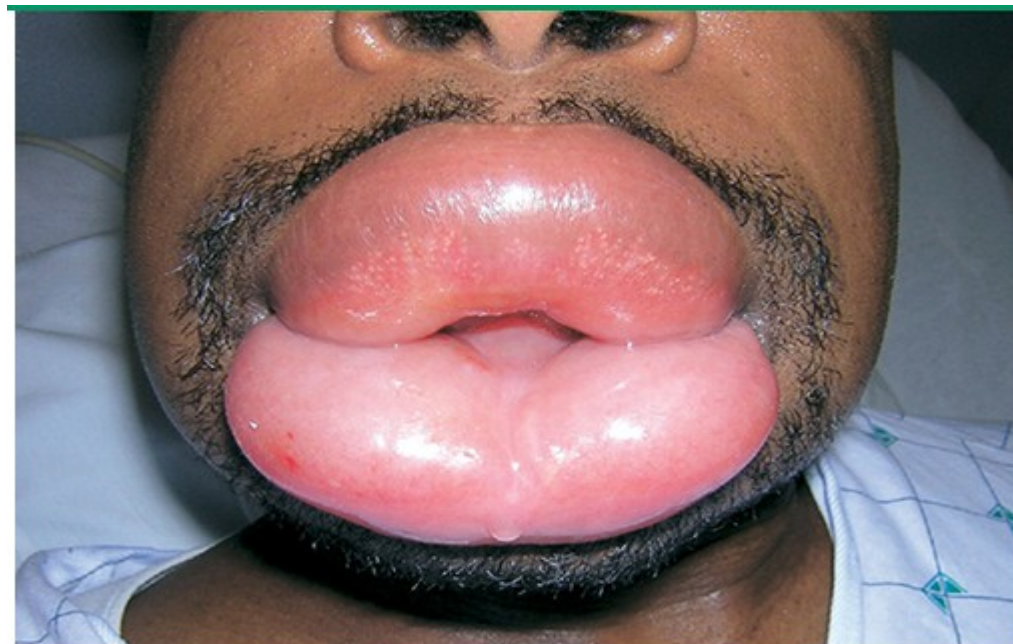
❖MOA

- ✓ Inhibit the converting enzyme (peptidyl dipeptidase) that hydrolyzes angiotensin I to angiotensin II and inactivates **bradykinin**, (**a potent vasodilator**)
- ✓ The enzyme is also responsible for **inactivation of bradykinin**
- ✓ blood pressure principally by decreasing peripheral vascular resistance. **Cardiac output and heart rate are not significantly change**

❖Drugs in this classes are:-

- captopril, Enalapril (pro-drug) , **Lisinopril**, Benazepril, fosinopril, moexipril, perindopril, quinapril, ramipril, and trandolapril

S/E:



Pharmacokinetics & Dosage

Enalaprilat,

- Peak time = 3-4hrs
- T_{1/2} = 11hrs
- Dose = 10–20 mg once or twice daily

➤ Lisinopril

- T_{1/2} = 12hrs
- Dose = 10–80 mg once daily

- All of the ACE inhibitors except **fosinopril and moexipril** are eliminated primarily by the kidneys

Toxicities

✓ Hypotension

- ✓ acute renal failure
- ✓ hyperkalemia,
- ✓ dry cough

C/I

- ✓ Second and third trimester Pregnancy because of fetal **anuria** and **hypotension**

Angiotensin Receptor-blocking

MoA

➤ complete inhibition of angiotensin action (greater than ACE- inhibitors)

Agents losartan, valsartan, azilsartan, candesartan, eprosartan, irbesartan, olmesartan and telmisartan

Toxicities

- Similar to ACE- Inhibitors with low dry cough effect